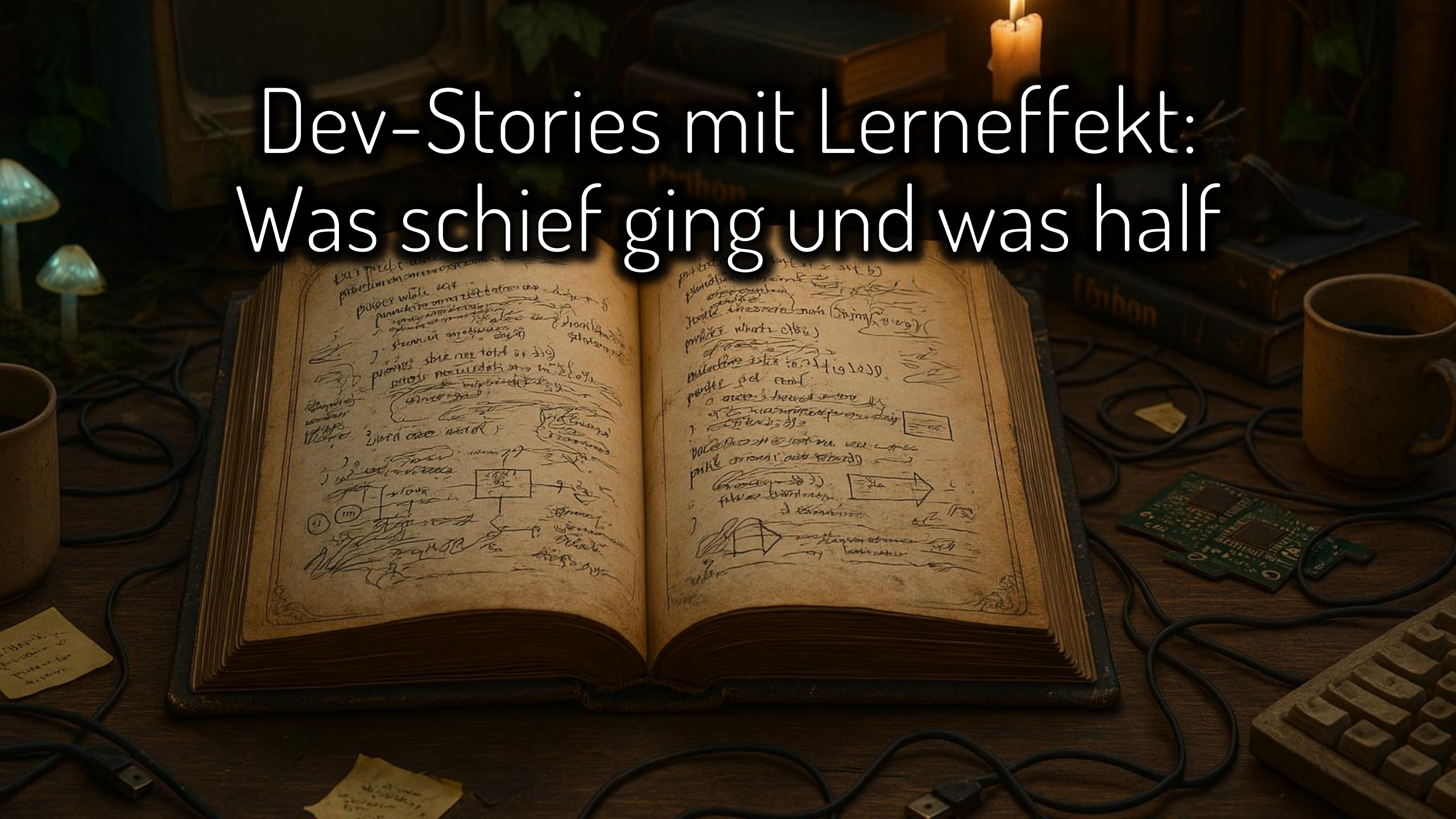


Dev-Stories mit Lerneffekt: Was schief ging und was half



Johannes Rabauer

Work Experience

 Nuclear Power Plant

 Radar Station

 Institute for Information Systems (iisys)

Senior Software Engineer

XDEV Software GmbH

 Helping out customers in projects

 Open source contributor



Story Time

Use IDE formatting profiles (IntelliJ, Eclipse, VS Code), enforce via CI (e.g., Maven build fails on style errors), and auto-format during build.

Static code analysis increases code quality, security, and performance. Tools like PMD and OWASP help identify issues early. GitHub Dependabot and Renovate ensure dependencies stay updated, avoiding migration issues.

Tests prevent regressions and document edge cases. Especially valuable in large, unfamiliar codebases. Always write tests, even for small functions. UI tests can be improved via TCI. Simplify to encourage clients not to disable them.

Dockerizing development environments eliminates flaky tests caused by inconsistent local setups. Use Docker and Docker Compose for reproducible builds across OS platforms.

Using standard solutions reduces complexity, bugs, and unknown edge cases. Avoid building custom systems for common problems. In one case, hundreds of lines in C# were replaced with 10 lines of Java by using a standard library.

If no standard exists, open source your solution. Examples include xdev-software's public repositories. This encourages community feedback and occasional contributions. Some components were even adopted by the Vaadin framework.

Standardizing workflows avoids knowledge silos and enables team continuity. Use Git workflows (feature branches), issue trackers (Jira, GitHub), and Wikis (Confluence) to maintain clarity and shared ownership. Monitoring and alerting (e.g., Grafana, Prometheus) detect system issues like memory leaks before failures occur.

Apply changes incrementally. Small, frequent changes are less likely to be rejected. Use commit cards to guide improvements.

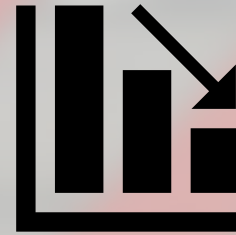
Rotate people in projects to spread knowledge, increase resilience, and challenge stagnant routines. Raises the bus factor and refreshes perspectives.

The Bus Factor





High Bus Factor



Low Bus Factor



Software development



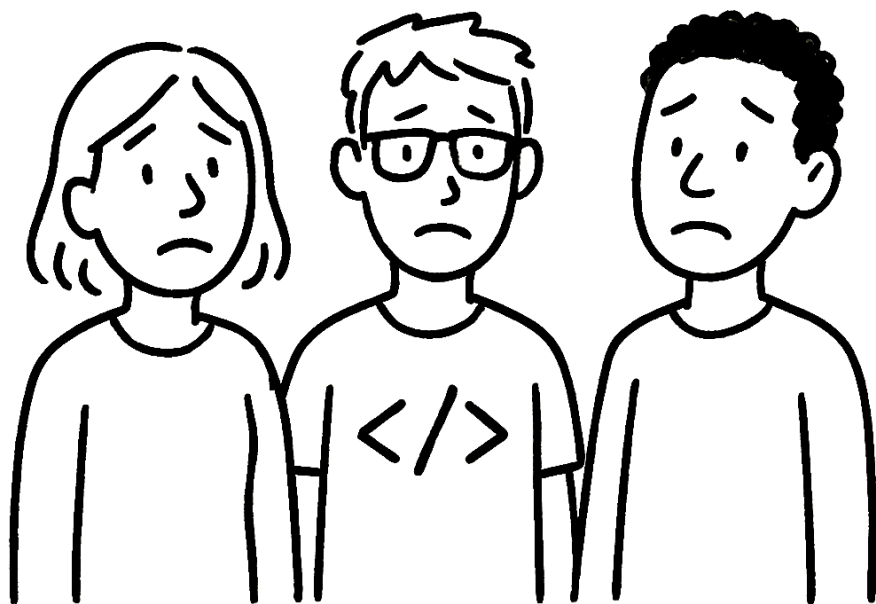
Software development



before 2010



after 2010



Robert
(Life Exp. 56y)



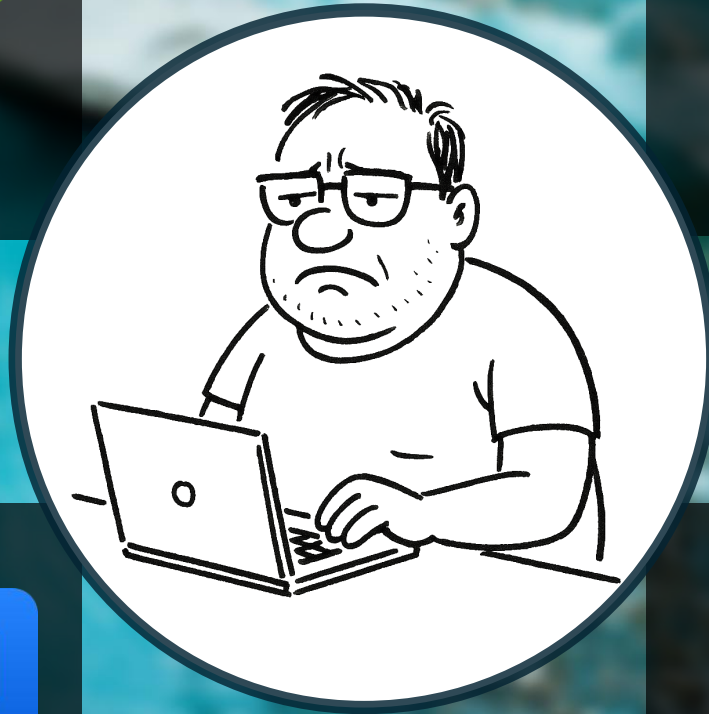
Standardize Workflow



Git-Host



Git feature
branch workflow



Issue Tracker



Wiki

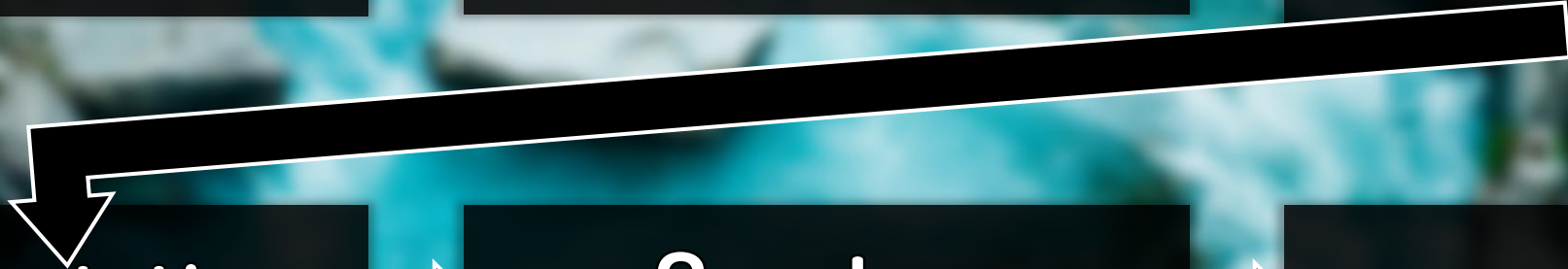
Create Issue



Create
Branch



Do the work



Documentation
(Issue and Wiki)



Create a
Pull Request



Peer Review



Merge



Close Issue



**Knowledge &
Quality**



Independence



Documentation

How it
started



Robert
(Life Exp. 56y)

How it's
going



Robert
(Life Exp. 84y)

A fight of the mind



Each call of a Stream APIs in a new line

```
List<Card> heartCards = deck.stream().filter(card -> card.suit.equals("♥")).toList();
```

```
List<Card> heartCards = deck  
    .stream()  
    .filter(card -> card.suit.equals("♥"))  
    .toList();
```



All calls

one line

```
long cardAmount = deck  
    .stream()  
    .count();
```

```
long cardAmount = deck.stream().count();
```

Tabs vs. Spaces

{ } in new line?



Final Variables

How many
chars per line?

A close-up photograph of two sloths against a white background. One sloth is in the foreground, its head tilted upwards with its mouth open, showing its teeth. Another sloth is behind it, also looking up. A semi-transparent dark grey horizontal band is positioned across the middle of the image, containing the text 'Code Formatter' in a bold, black, sans-serif font.

Code Formatter

Settings

format

×

Editor > Code Style > Java

Revert changes

←

→

Appearance & Behavior

Keymap

Editor

Code Style

Java

Groovy

Shell Script

SQL

General

SQL:2016, Generic

Apache Derby

Db2

H2

HSQLDB

MS SQL Server, MS Azure

MySQL, MariaDB

Oracle

PostgreSQL, Greenplum, Redshift

SQLite

Sybase ASE

Style Sheets

CSS

Less

SCSS

Inspections

Live Templates

Scheme: Default IDE

⚙

Copy to Project...

Duplicate...

Restore Defaults

Export

Import Scheme

IntelliJ IDEA code style XML

Eclipse XML Profile

Tabs and Indents

Spaces

Wrapping

Hard wrap at:

Wrap on typing

Visual guides

Keep when reformatting

Line breaks

Comment at first column

Control statement in one line

Multiple expressions in one line

Simple blocks in one line

Simple methods in one line

Simple lambdas in one line

Simple classes in one line

Ensure right margin is not exceeded

Braces placement

In class declaration

In method declaration

In lambda declaration

Other

Extends/implements/permits list

Align when multiline

Extends/implements/permits keyword

Throws list

Align when multiline

Set from...

Names

JavaDoc

Imports

Arrangement

Code Generation

Java EE Names

```
do x++; while (x < 10000);
switch (a) {
case 0:
break;
case 2:
case 3:
return;
default:
doDefault();
}
try (MyResource r1 = getResource(); MyResource r2 = null) {
doSomething();
} catch (Exception e) {
processException(e);
} finally {
processFinally();
}
do {
x--;
} while (x > 10);
try (MyResource r1 = getResource();
MyResource r2 = null) {
doSomething();
```

?

OK

Cancel

Apply



Any style is fine



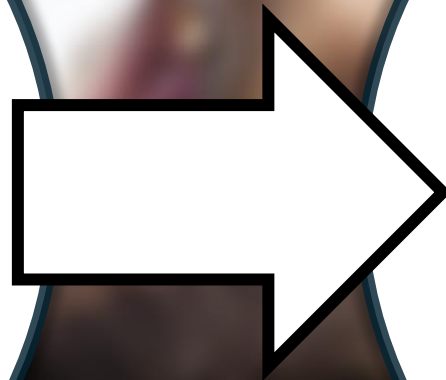
Keep focus



Automation

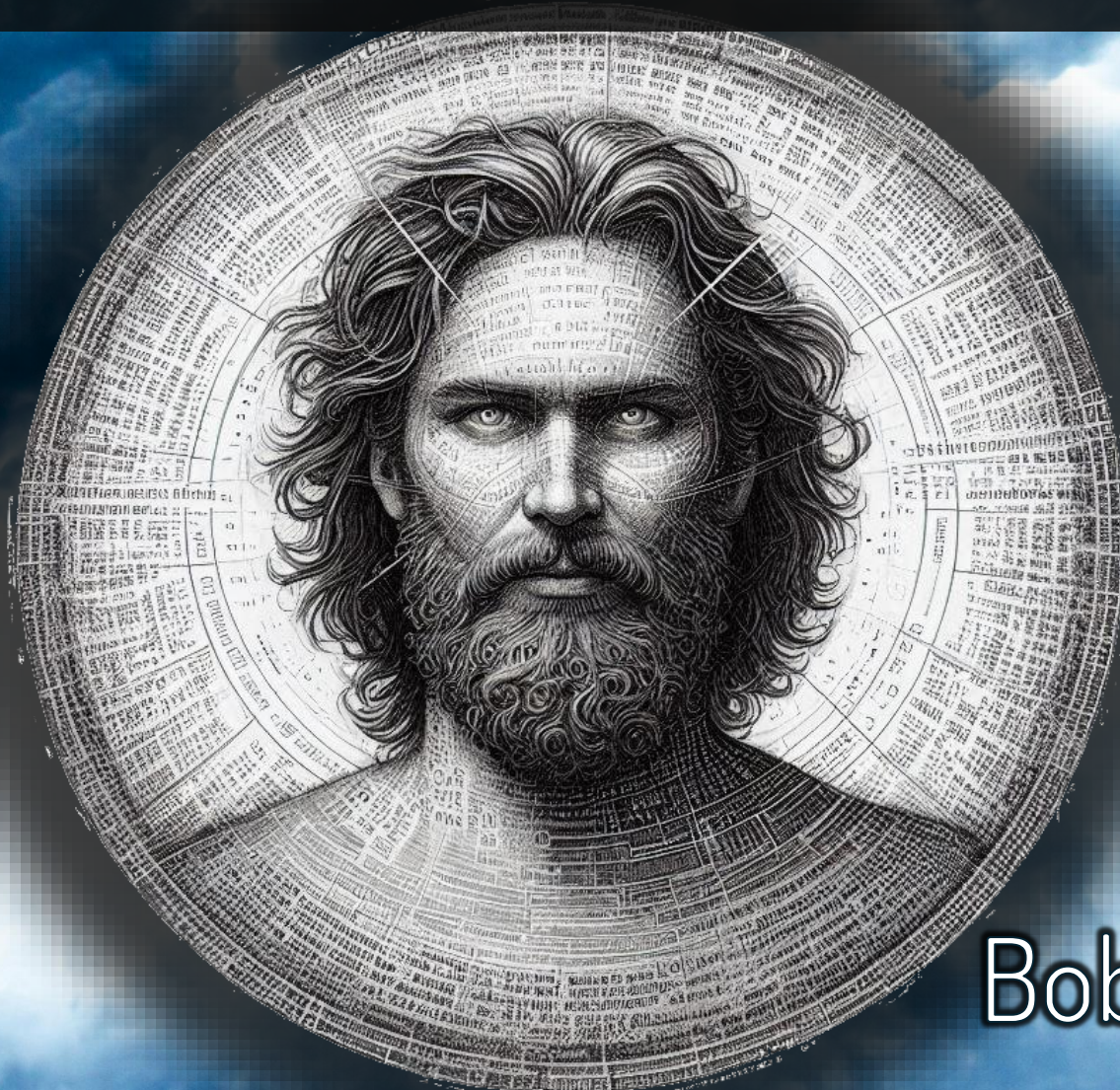


Juanita
(Life Exp. 68y)



Juanita
(Life Exp. 89y)

The Perfect Programmer?



Bob



D.U.T.Y.



D.U.T.Y.

Dependency Update Tools for You



D.U.T.Y.

Dependency Update Tools for You



Search dependencies



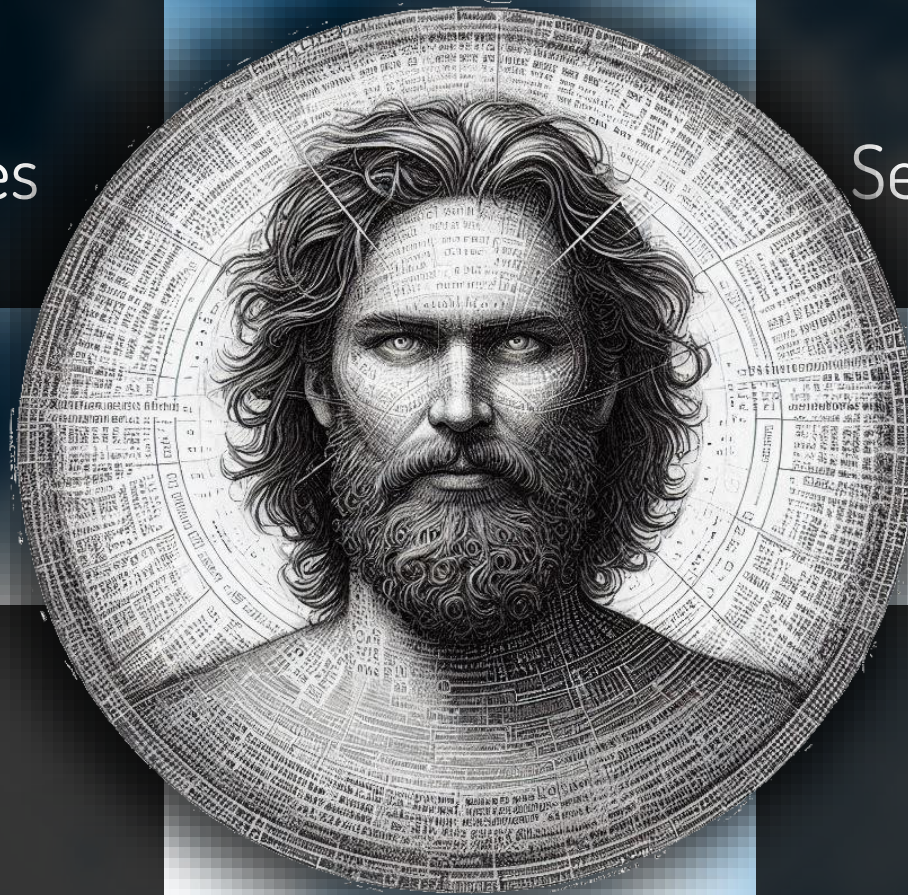
Search newest versions



Update dependencies



Create Pull Requests

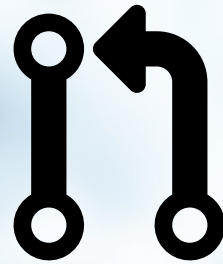




<https://github.com/JohannesRabauer/dependabot-demo>



Up to date



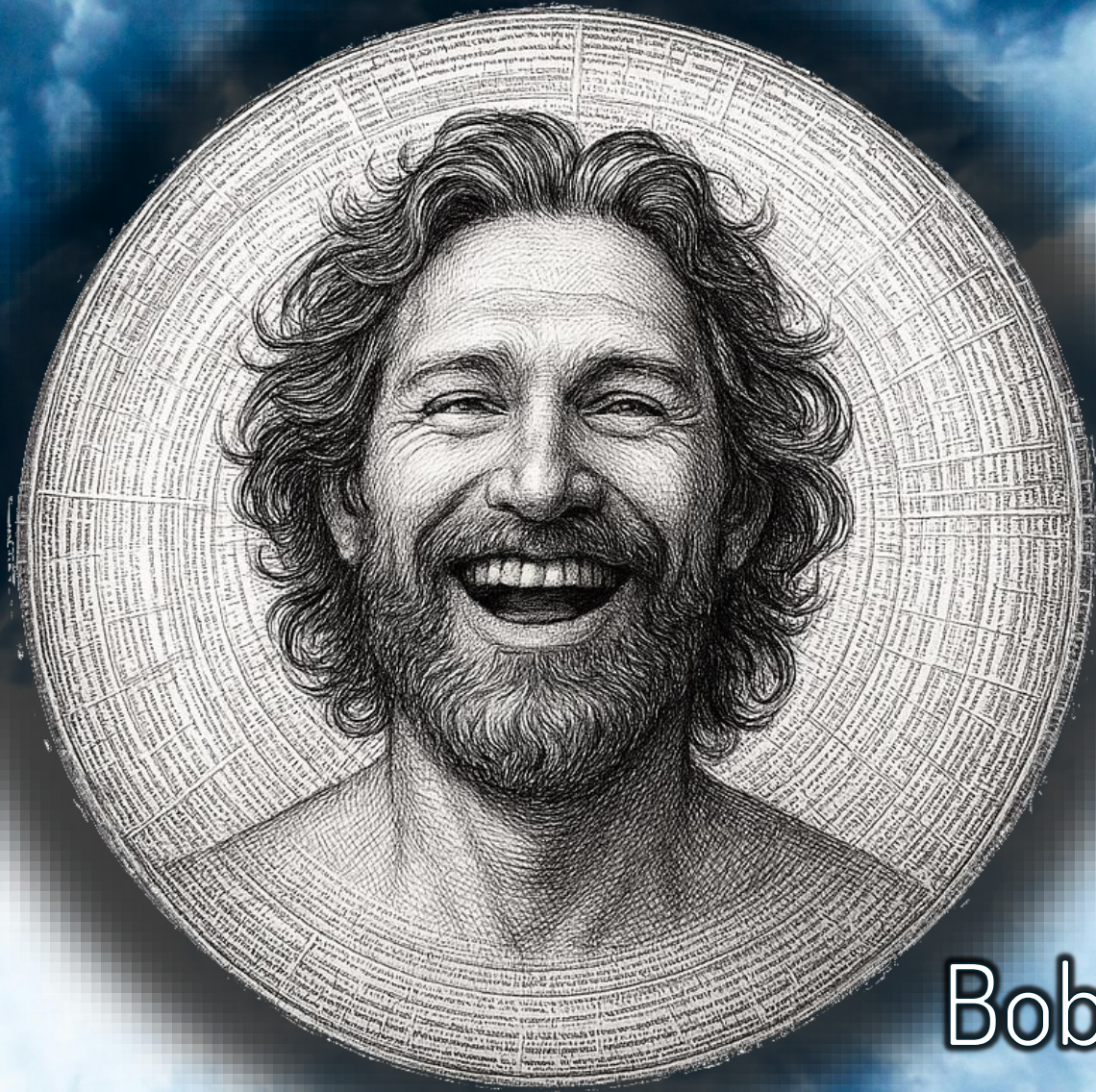
One click merge



Tests



<https://github.com/JohannesRabauer/dependabot-demo>



Bob

Johannes



*"We've got a long way to go
and a short time to get there"*

– Jerry Reed





Test EVERYTHING

If you find weird behavior

```
/**
 * TODO: Could we change this in the future?
 */
@Test
void nullCardShouldNotBeGreaterThanRealCard() {
    Card realCard = new Card(10, "♥");
    Card nullCard = null;

    int result = CardComparator.compare(nullCard, realCard);
    assertTrue(result > 0);
}
```

If you know what behavior you want

```
@Test
void nineCardShouldBeSmallerThanTenCard() {
    Card card10 = new Card(10, "♥");
    Card card9 = new Card(9, "♥");

    int result = CardComparator.compare(card9, card10);
    assertTrue(result < 0);
}
```

If you don't know if something behaves correct

```
/**
 * TODO: This might be correct, or it might be a bug.
 * Either way, here is the test :)
 */
@Test
void heartCardShouldBeGreaterThanSpadeCard() {
    Card heartCard = new Card(10, "♥");
    Card spadeCard = new Card(10, "♠");

    int result = CardComparator.compare(heartCard, spadeCard);
    assertTrue(result > 0);
}
```



Documentation



**Fails when
behavior changes**

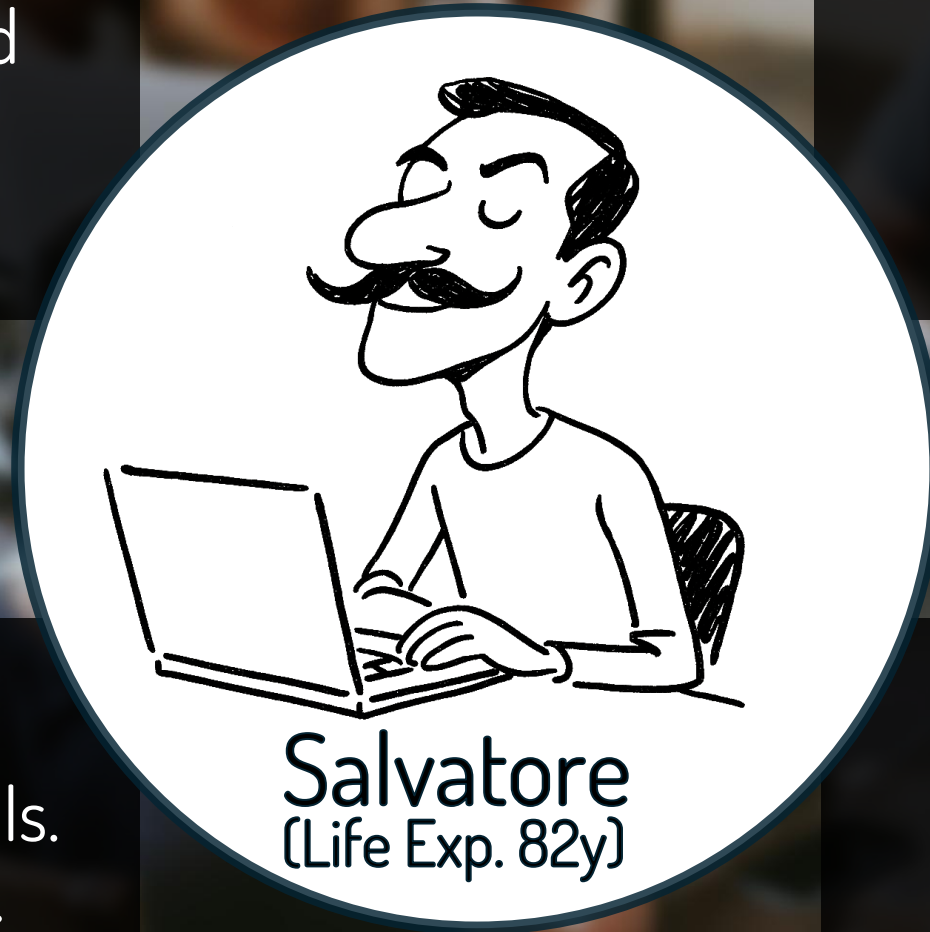


Write tests



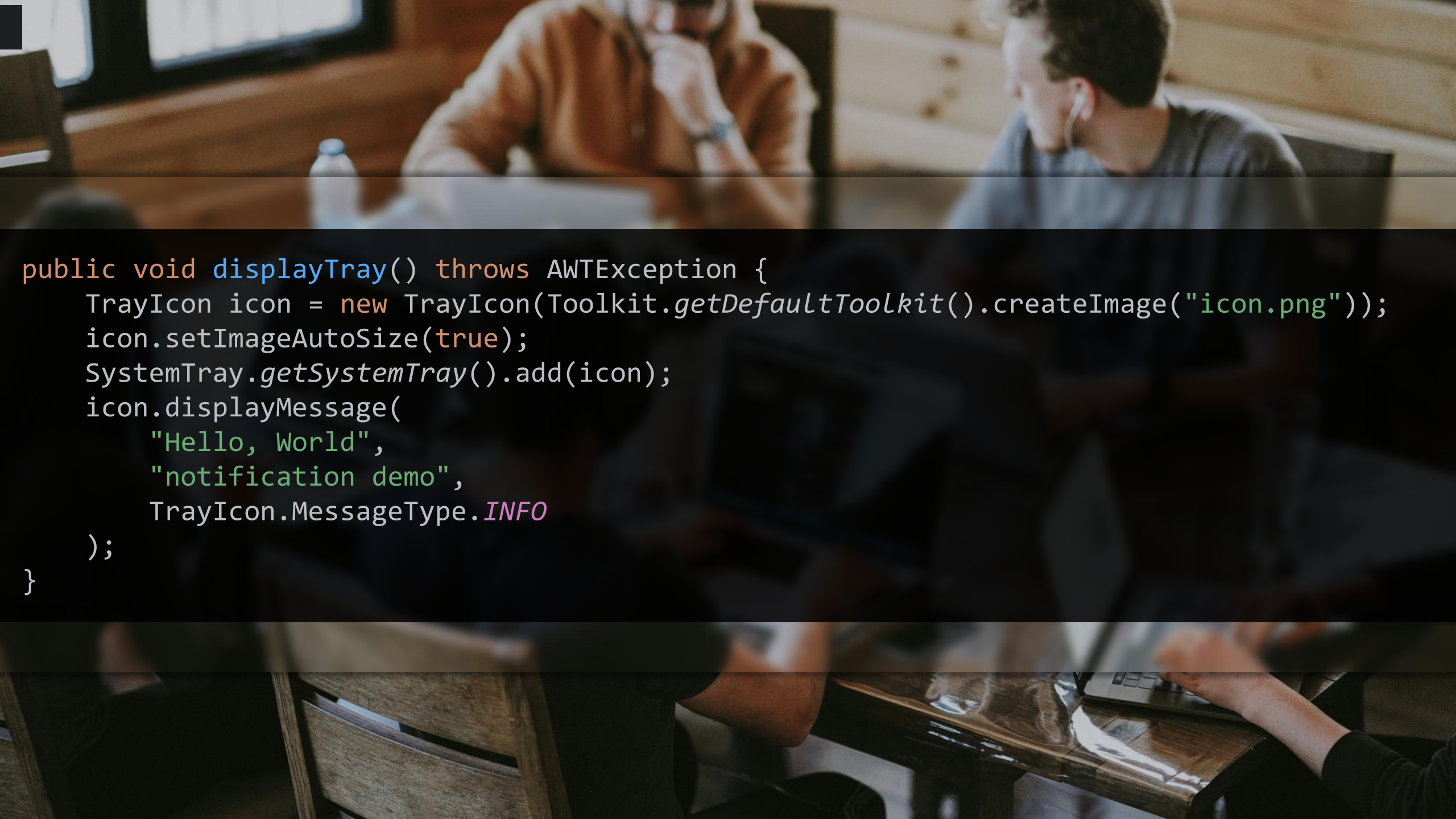
Follow your ideas and
dreams

Don't mind your
colleagues' opinions



Don't use standard tools.
Only you are perfect.

If you want
something done right,
do it yourself



```
public void displayTray() throws AWTException {  
    TrayIcon icon = new TrayIcon(Toolkit.getDefaultToolkit().createImage("icon.png"));  
    icon.setImageAutoSize(true);  
    SystemTray.getSystemTray().add(icon);  
    icon.displayMessage(  
        "Hello, World",  
        "notification demo",  
        TrayIcon.MessageType.INFO  
    );  
}
```

~~Follow your ideas and dreams~~
but stay practical
and efficient



Salvatore
(Life Exp. 82y)

Value
~~Don't mind~~ your
colleagues' opinions
and ask them for
help or advise

~~Don't~~ use standard tools.
~~Only you~~ are perfect.
Nobody is

If you want
something done right,
~~do it yourself~~
find existing solutions



Less Bugs



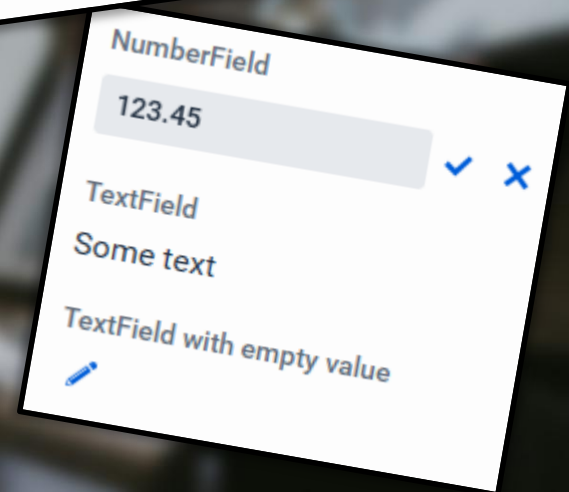
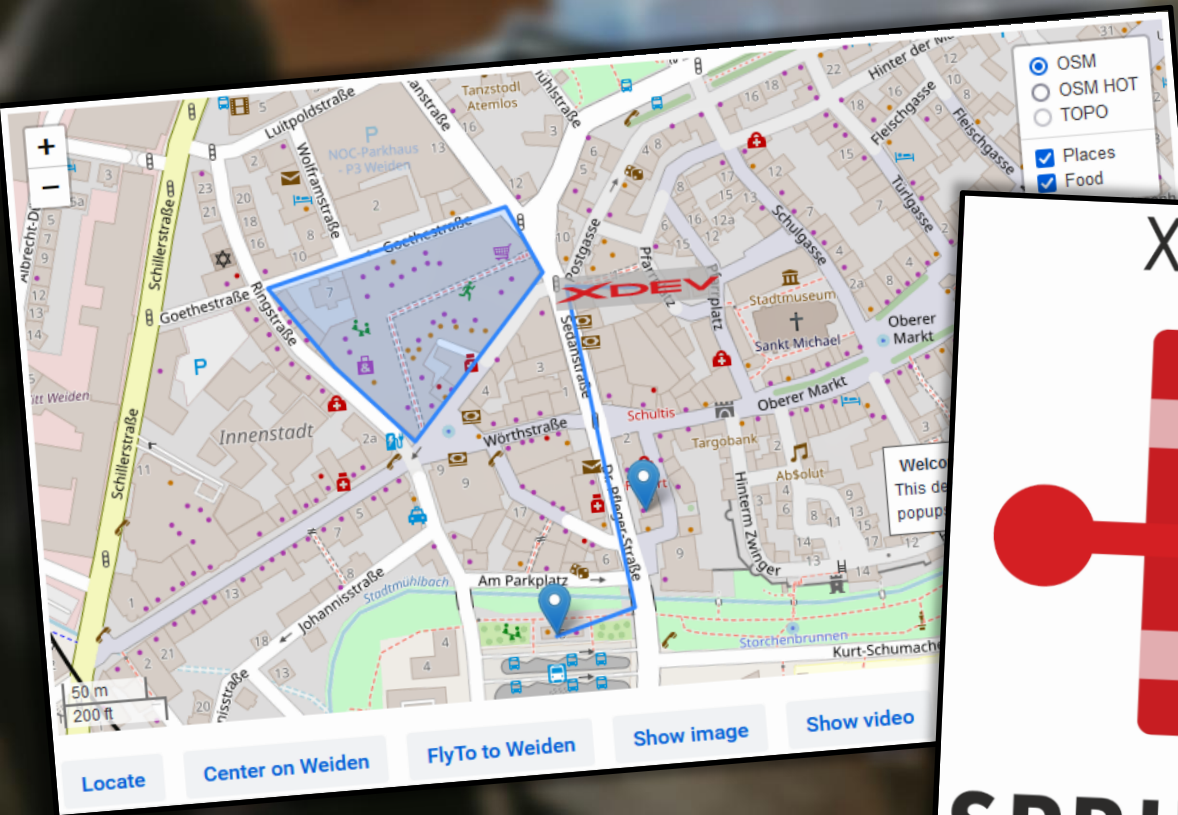
Less Duplicates



Less Complexity



No solution? Open source your own!





Community



Expertise



Feedback

The image shows a carousel with a highly decorative canopy. The canopy features ornate, colorful patterns in shades of gold, red, yellow, and blue. A central, multi-tiered decorative finial is visible. Below the canopy, several carousel horses are visible, each with unique paint schemes in colors like white, red, purple, blue, and green. The horses are mounted on a blue metal frame. The background is filled with lush green trees.

Rotate Responsibilities



Bus Factor



Learn



Ask for Help



A young child with blonde hair, wearing a white t-shirt and blue shorts, is walking away from the camera on a paved street. The child is holding a black leash attached to a small, white dog with brown spots. To the left of the child are large wooden barrels. In the background, there are some red flowers in a planter. The scene is captured in a soft, slightly blurred style.

Baby Steps



Baby Steps



Unity



Steady Growth

Danke!

Freue mich über Feedback



Danke!

Freue mich über Feedback



XDEV Commit Cards

Don't reinvent
the wheel



Is there a library that does
exactly what you need?
If there is already maintained
and secure code, why
shouldn't you use it?

Write tests



Commit
Cards

software